

B30801 Managerial Accounting Autumn 2025

Doug Skinner

**Session #3: Why Allocate Costs?
Activity-Based Costing (ABC).**

CHICAGO BOOTH



The University of Chicago Booth School of Business

VMD key-takeaways:

- Business changed such that allocation base (DL\$) did an increasingly poor job of allocating overhead costs to different products (X-rays vs. MRIs).
- Because VMD used full cost to price products, prices set too low (too high) for MRIs (x-rays) products, which led to *demand-driven* death spiral.
- Use of additional cost pools helped, but allocations never perfect – no “true, accurate” cost (allocations *inherently arbitrary*).
- Controllability and buy-in issues create incentive problems (e.g., Jankoski).

Indirect costs

- Direct costs can be directly and unambiguously assigned to cost objects. All other costs are indirect.
- Indirect costs are of two types:
 - **Joint costs:** “cost of a production process that yields multiple products simultaneously.”
 - **Common (shared) costs:** “cost of operating a facility, activity, or cost object when (it) is shared by two or more users” (e.g., HR, IT or other support services).



Other examples; chicken + eggs, beef + milk + leather,

Refining crude oil simultaneously produces:

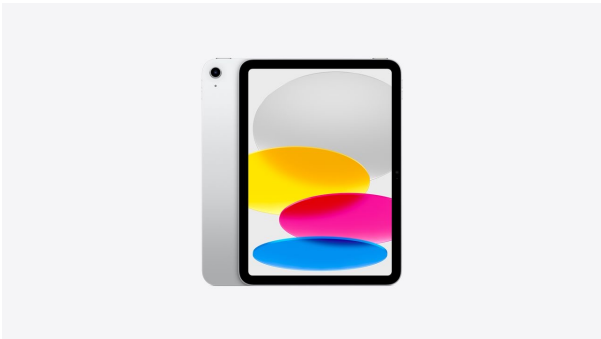
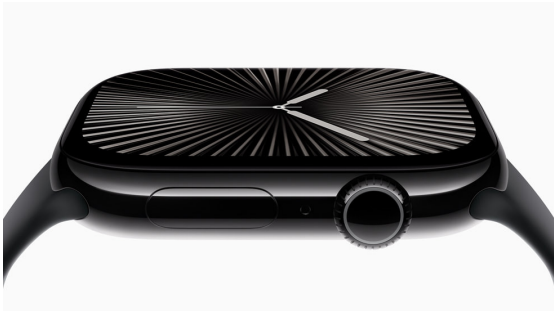
- Gasoline
- Jet fuel
- Diesel fuels
- Lubricating oils, asphalt, etc.

The cost of crude oil is a *joint cost*—there is no obvious way of allocating this cost to the products.

The brand apple itself. very costly to develop and maintain. how to allocate these brand building costs? because it supports a multitude of many products that benefit from a strong apple brand and its iOS.







Why allocate indirect costs?



This is one (very good) reason for cost allocation – basic microeconomics: if the price of a good is lower than its (opportunity) cost, the consumer will demand “too much”. Here allocating *some* of the cost to the consumer increases the cost he internalizes and so lowers the amount he demands.

For many indirect (overhead, joint, or common) costs, there is no natural underlying activity measure that drives cost and so no obvious allocation mechanism – as a result, there *can be no “true” or “correct” (allocated) cost*. Indirect cost allocation is *inherently arbitrary*.

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→ Why allocate costs at all?

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→ Why allocate costs at all?

Consider marketing, a big unit with substantial costs that produces overall brand marketing and advertising.

1. Should we allocate the cost of marketing to underlying business units?
2. If so, how?

Booth verticals:

- FT MBA
- Part-time MBA (evening, weekend)
- EMBA
 - London, Chicago, Hong Kong campus ops.
- One-year masters (MiM, MiF)
- Ph.D.
- Executive education

Each benefits from the Booth brand and in addition has its own advertising budget.

Do we allocate the common costs of investments in brand to these business units?

Why allocate indirect Marketing costs?

- If no costs allocated to business unit P&L, business unit leaders demand “too much” marketing because the cost (to them) is effectively zero.
- In the long-run, the entity needs to recover all costs, which includes indirect marketing costs.

→ *Internally, there is no obvious way to measure marketing output (what is the activity measure?) and so no obvious way of allocating costs or otherwise “pricing” its services.*

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- If we believe external market is competitive, use *market prices* to price marketing services internally (a transfer price).

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- Problem: if we set the (internal) price too high, business units will purchase from outside rather than internally, resulting in underutilization of the marketing unit for which costs are largely fixed and unavoidable.

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This means we need to be able to set the price at the "right" level – larger than zero but no higher than market price.

1. **External financial reporting and tax rules.** GAAP rules (and tax law) *require* absorption costing for manufacturing costs. As we have already seen, we would not normally use these “fully loaded” costs to make internal business/management decisions.
 - The organization keeps two sets of books, one for external reporting and another for internal purposes—management decision-making and control (i.e., “internal view management P&Ls” and “external view P&Ls”). But that’s costly and can be confusing.



Cronos Group Inc.
Consolidated Statements of Financial Position
As at December 31, 2018 and December 31, 2017
(in thousands of CDN \$)

	Notes	2018	2017
Assets			
Current assets			
Cash	25(a)	\$ 32,634	\$ 9,208
Accounts receivable	25(a)	4,163	1,140
Sales taxes receivable		3,419	3,114
Prepays and other receivables	25(a)	3,876	790
Biological assets	7(a)	9,074	3,722
Inventory	7(b)	11,584	8,416
Loan receivable	8,25(a)	314	314
Total current assets		65,064	26,704
Advances to joint ventures	9,25(a)	6,941	-
Investments in equity accounted investees	10	3,492	3,807
Other investments	11,25(c)	705	1,347
Property, plant and equipment	12	171,891	56,172
Intangible assets	6(a),13(a)	11,234	11,207
Goodwill	13(b)	1,792	1,792
Total assets		\$ 261,119	\$ 101,029

7. Biological assets and inventory

(a) Biological assets

The Company's biological assets consist of cannabis plants. The changes in the carrying amounts of the biological assets are as follows:

	2018 (Note 6)	2017 (Note 6)
Biological assets - beginning of year	\$ 3,722	\$ 1,795
Capitalization of production costs	7,145	3,983
Unrealized change in fair value of biological assets	11,568	7,637
Transferred to inventory upon harvest	(13,361)	(9,693)
Biological assets - end of year	<u>\$ 9,074</u>	<u>\$ 3,722</u>

As of December 31, 2018, it is expected that the Company's biological assets will ultimately yield approximately 6,303 kg of dry cannabis (2017 - 1,695 kg). As at December 31, 2018, the Company has 46,004 plants (2017 - 7,353 plants) classified as biological assets.

The Company measures its biological assets at fair value less costs to sell. This valuation is based on the expected harvest yield (in grams) for plants currently being cultivated, adjusted for the expected selling price less post-harvest costs attributable to bringing a harvested gram of cannabis to a saleable condition and ultimate sale (on a per gram basis). The Company accretes the fair value of each cannabis plant on a straight-line basis over the expected growing cycle. As at December 31, 2018, the plants were on average 6 weeks (2017 - 7 weeks) into the growing cycle 37% complete (2017 - 46%) and were ascribed approximately 37% (2017 - 46%) of their expected fair value at harvest date.

(c) Direct and indirect cost allocations

The direct and indirect costs related to biological assets and inventory are allocated as follows. The allocation basis was consistent for the years ended December 31, 2018 and 2017, unless otherwise specified.

Nature of cost	Allocation basis
Consumables (insect control, fertilizers, soil)	100% allocated to production costs; incurred to support plant growth
Labour costs (including salaries and benefits)	Allocated based on job descriptions of various personnel; 30% allocated to processing costs; 40% allocated to production costs; 30% allocated to operating expenses (2017 - 20%; 70%; and 10%; respectively)
Supplies and small tools	80% allocated to production costs; 20% allocated to processing costs
Utilities	Allocated based on estimates of usage, 10% allocated to processing costs; 90% allocated to production costs
Property taxes, depreciation, security	Allocated based on estimates of square footage, 20% allocated to processing costs; 50% allocated to production costs; 30% allocated to operating expenses
Packaging costs	100% allocated to processing costs

2. Cost-plus or cost-reimbursement contracts. Contracts with government agencies (e.g., aerospace, health care, universities for research funding, etc.) **reimburse full cost** of providing goods and services. Leads to two incentive problems:

- Suppliers incentivized to over-allocate indirect costs to project (overcharge);
- Suppliers can pass on all costs and not incentivized to manage costs prudently.

Common contract type in some industries. For us, similar for VITAL and some government contracts. Also for payroll outsourcing.

AstraZeneca vaccine document shows limit of no-profit pledge

Company has right under contract to declare pandemic over by July 2021



AstraZeneca's new factory in Sodertalje, Sweden. The company, which is developing a vaccine with Oxford university, received hundreds of millions in public money to fast track development © Bloomberg via Getty Images

Donato Paolo Mancini in Rome YESTERDAY

AstraZeneca, which has **promised not to profit** from its Covid-19 vaccine “during the pandemic”, has the right to declare an end to the pandemic as soon as July 2021, according to an agreement with a manufacturer.

The UK pharmaceutical company, which is developing a vaccine candidate with Oxford university, has said it would **provide doses on a cost basis** for at least as long as the pandemic lasts.

AstraZeneca will provide its C-19 vaccine on a “cost basis.”

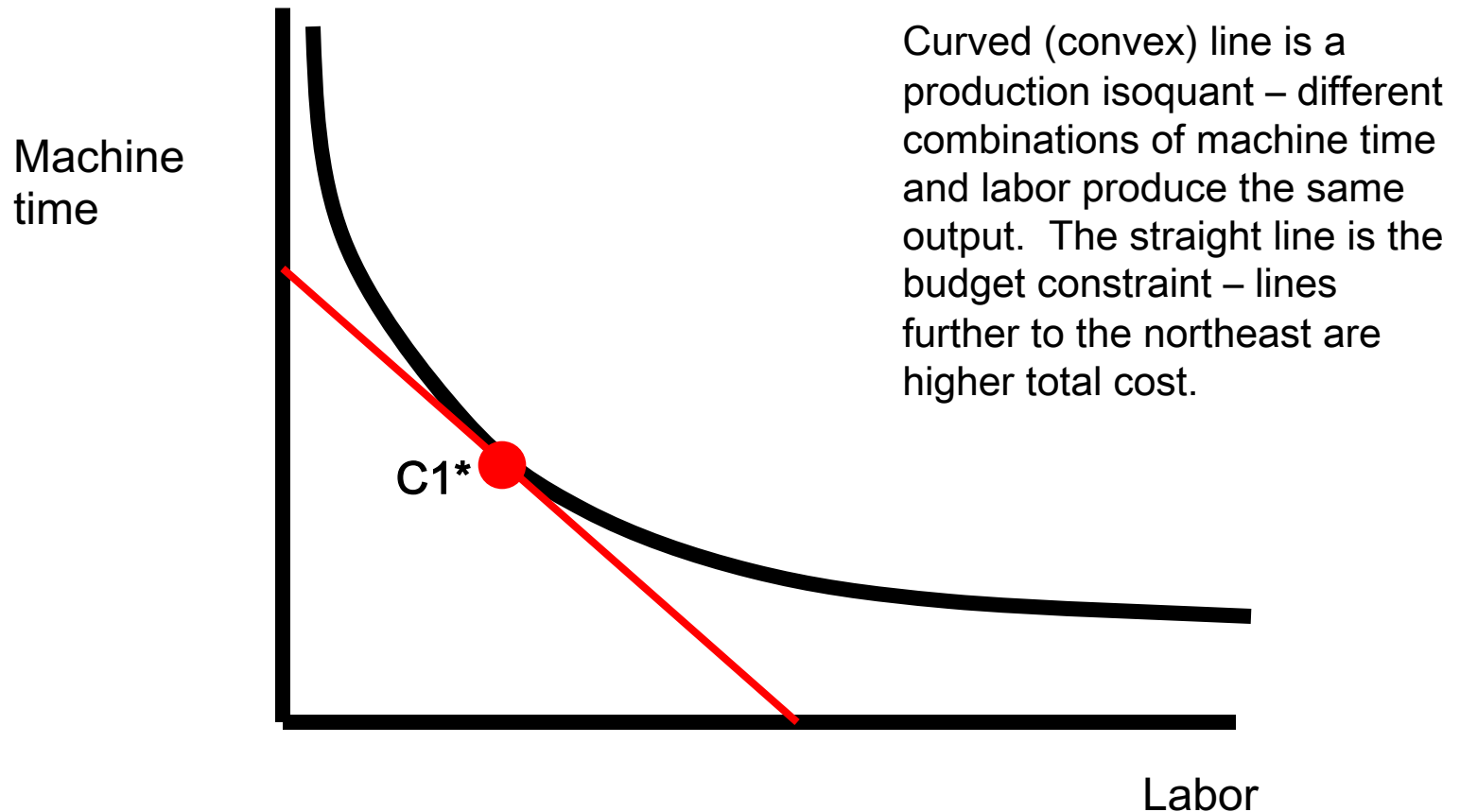
It would be interesting to know what this means – for example, since most of its costs are likely R&D costs, as opposed to direct manufacturing costs, it will make a big difference whether those costs are included and how they’re allocated to doses of the vaccine.

FT, 10.7.20

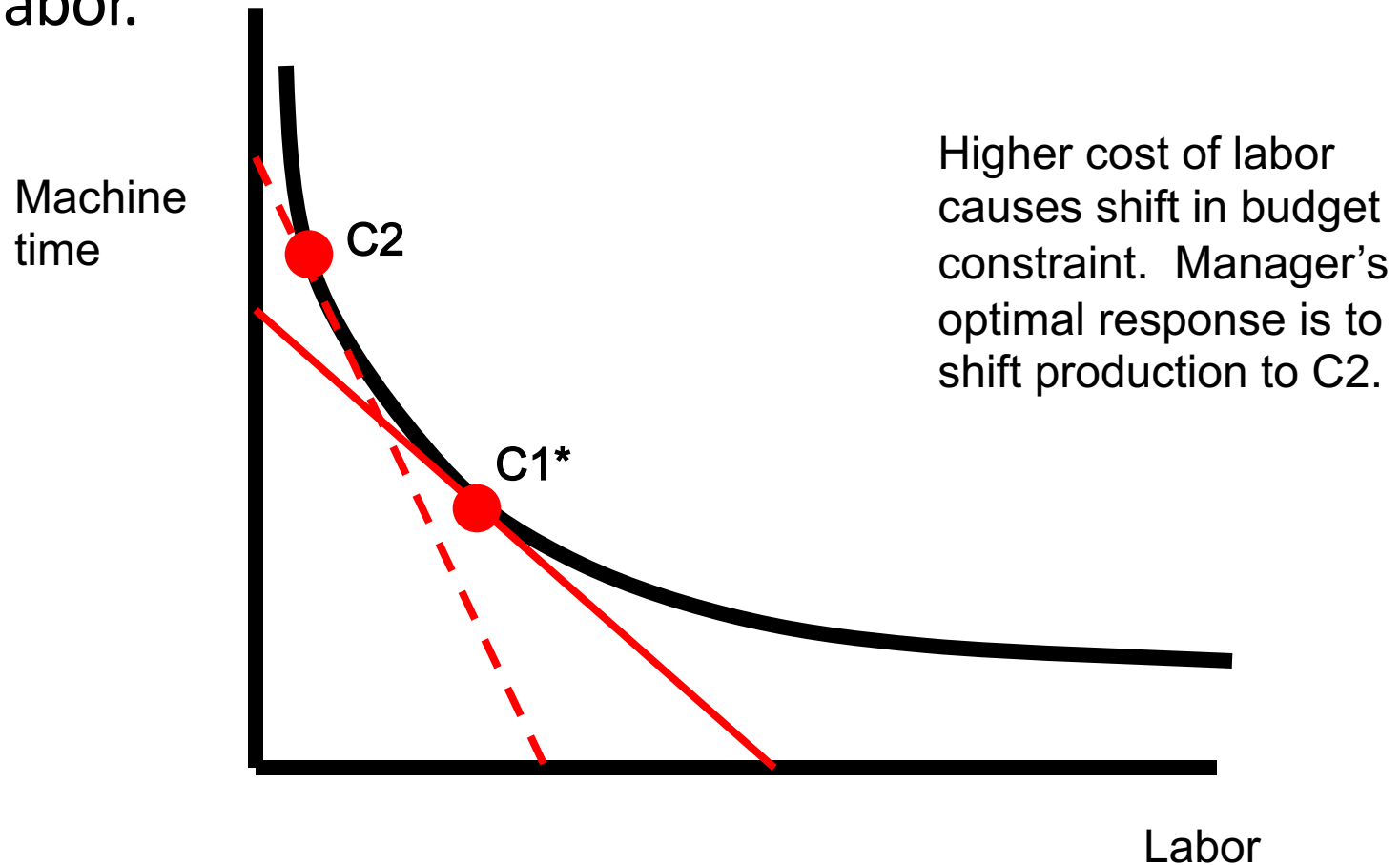
3. **Pricing decisions; Strategic decisions about products and product lines.** All costs must ultimately be recovered. Cost data *inform* (but does not *determine*) pricing.
- Beware the demand-induced death spiral (VMD) – if we over-allocate indirect costs to products we price ourselves out of the market. Conversely, if we price too low we will not recover all costs.
 - Demand and supply forces determines prices, not our costs.

-
4. **Decision-making and control.** Cost allocation is a key part of the Rules of the Game (e.g., it affects how internal decisions are made and how performance is measured – coordination and communication re resource allocation).
- Production decisions depend on how input prices are set within organizations.

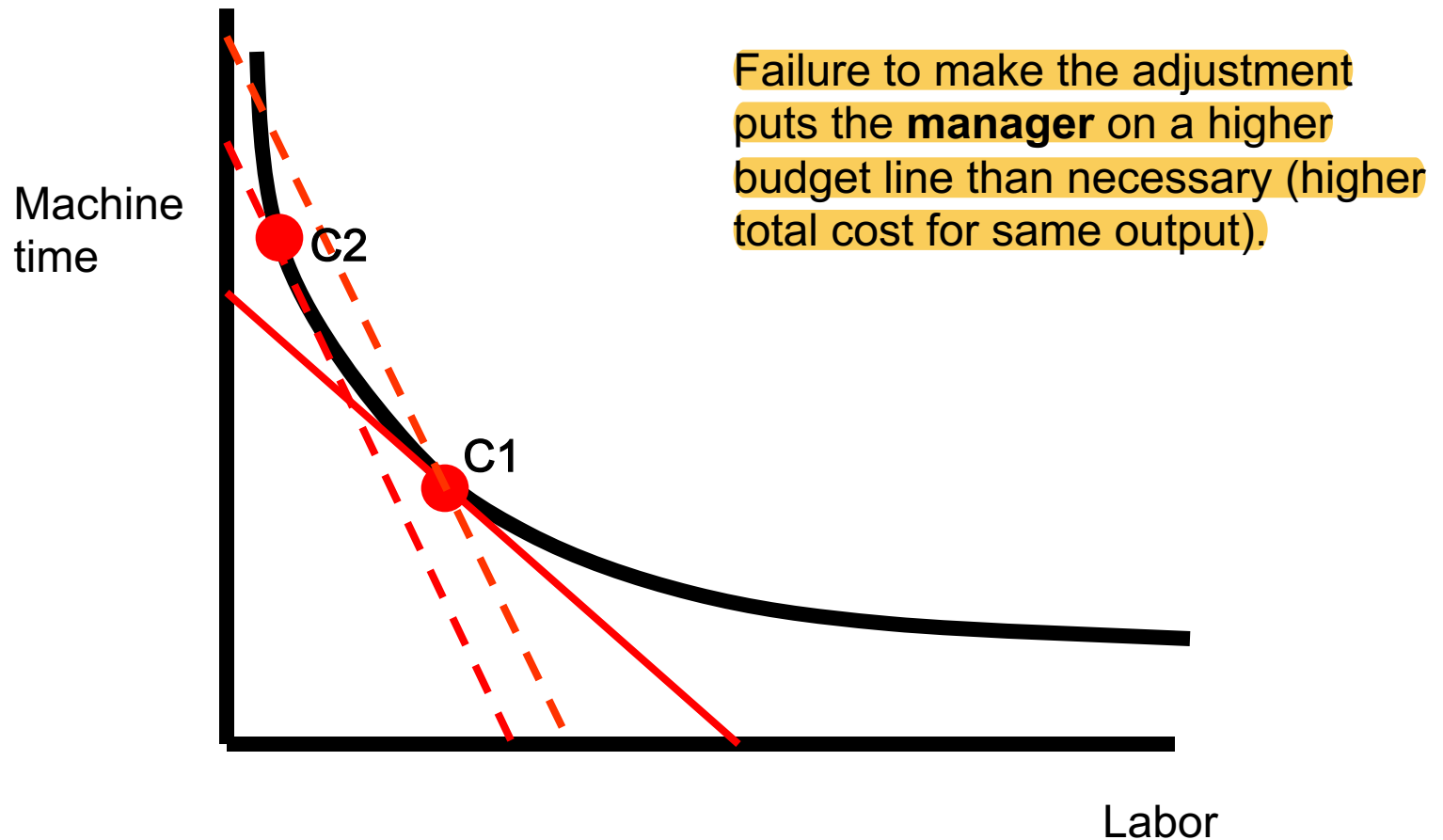
The optimal input mix ($C1^*$) minimizes cost for given level of output



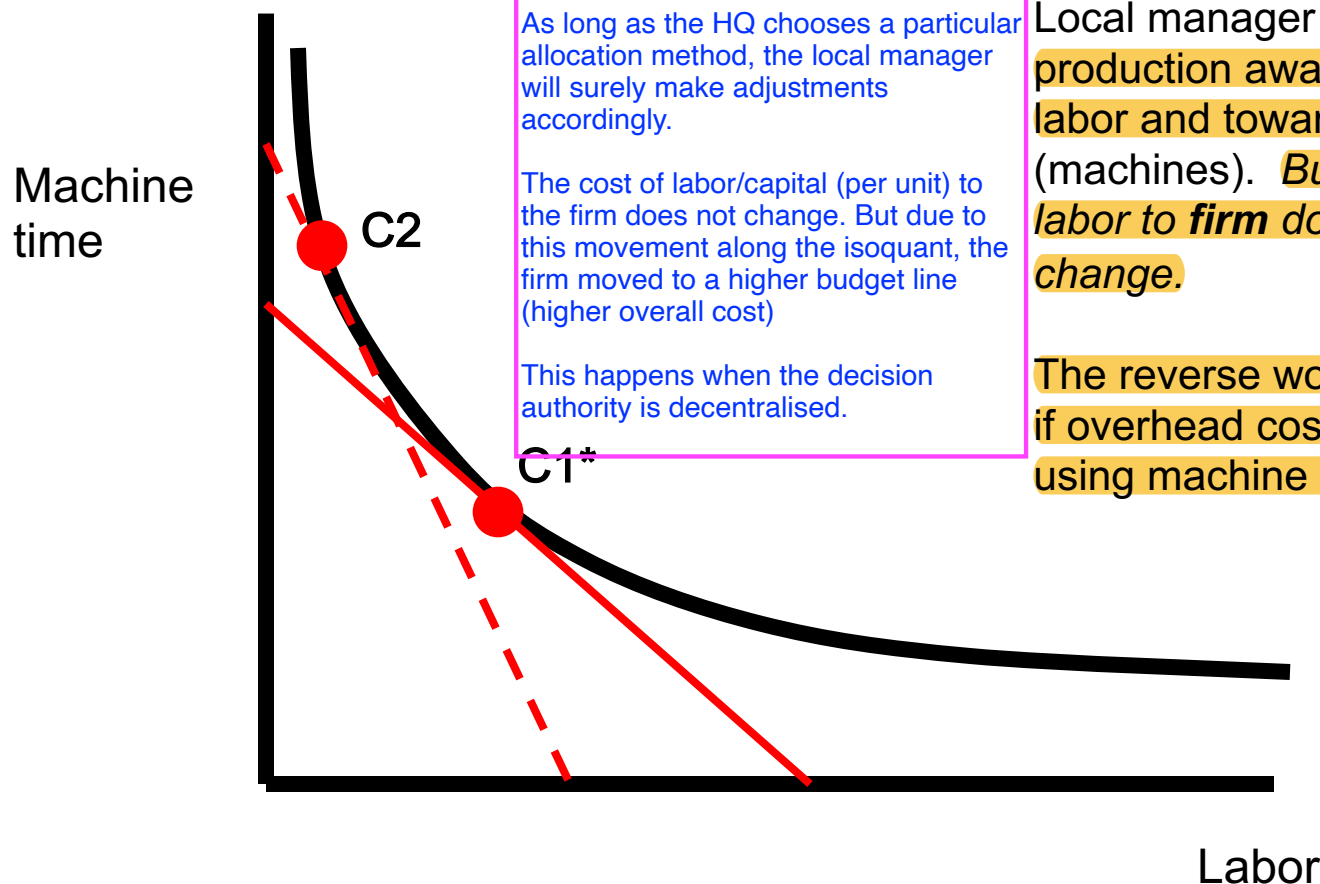
If the cost of labor increases, the manager substitutes machine time for labor.



The move to C2 is optimal.



When overhead is allocated based on direct labor hours, it is as-if the cost of labor increases.



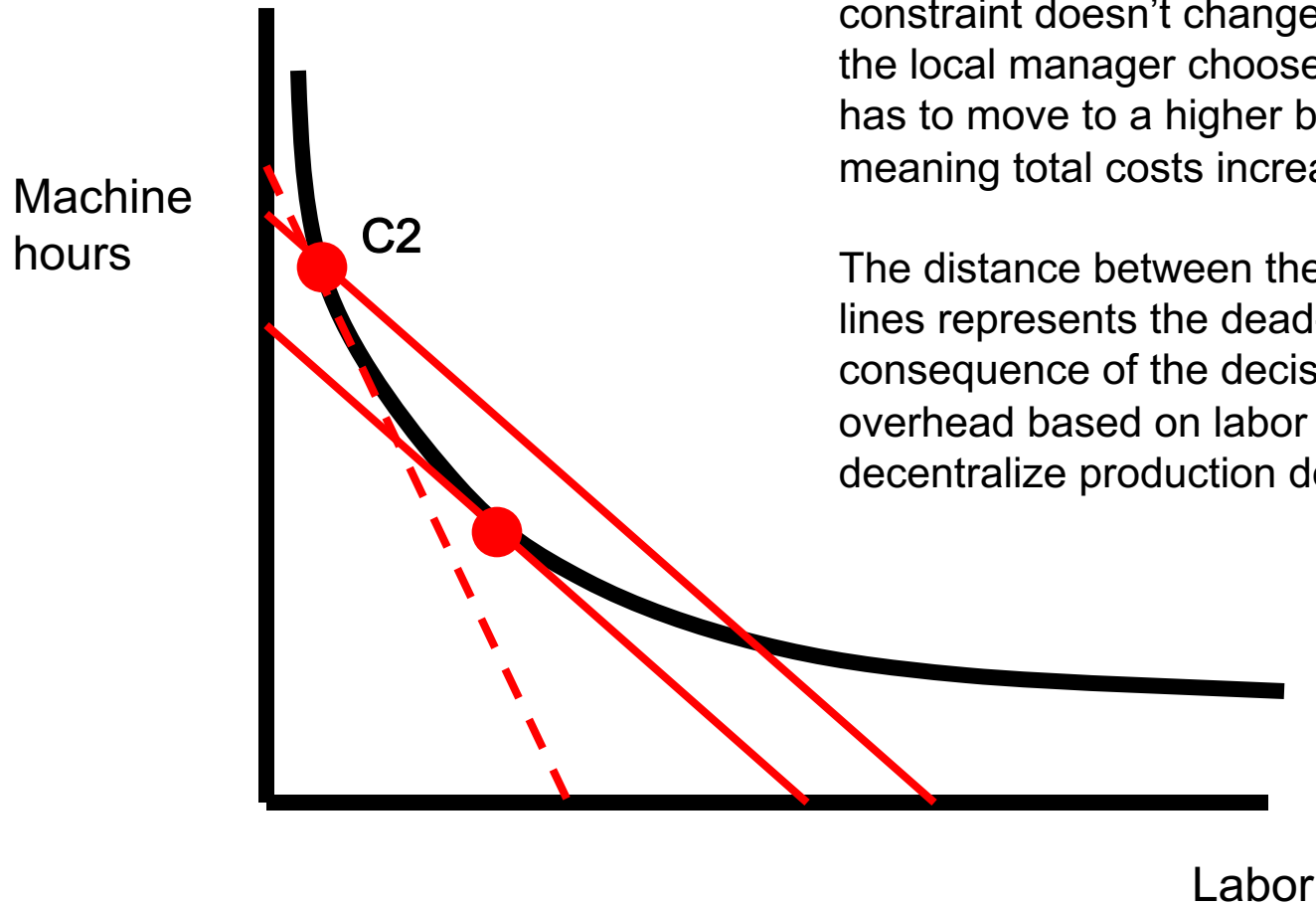
As long as the HQ chooses a particular allocation method, the local manager will surely make adjustments accordingly.

The cost of labor/capital (per unit) to the firm does not change. But due to this movement along the isoquant, the firm moved to a higher budget line (higher overall cost)

This happens when the decision authority is decentralised.

Local manager shifts production away from labor and towards capital (machines). But cost of labor to firm does not change.

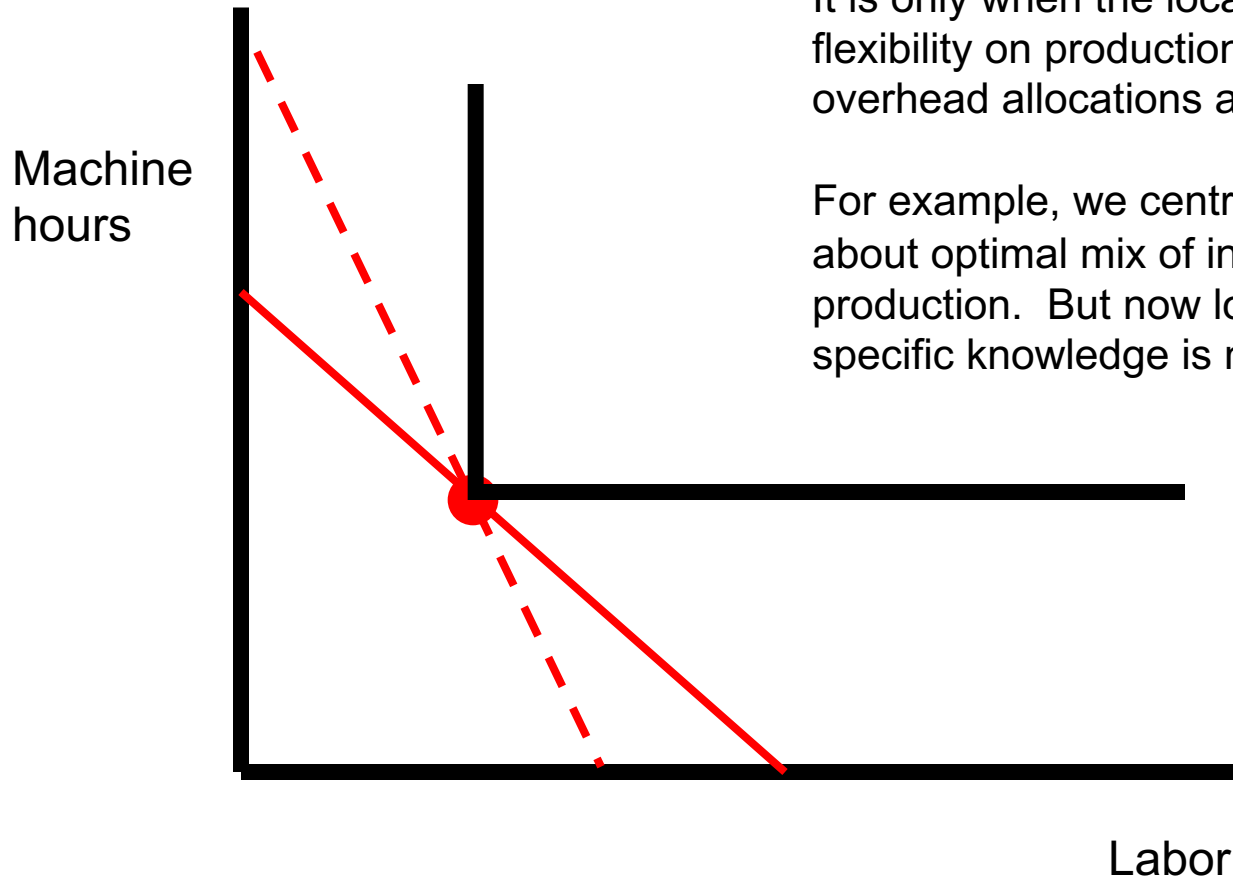
The reverse would happen if overhead cost allocated using machine hours.



From the firm's standpoint, prices have not changed, so the slope of the budget constraint doesn't change. But because the local manager chooses C2, the firm has to move to a higher budget line, meaning total costs increase.

The distance between the two solid lines represents the dead-weight loss, a consequence of the decision to allocate overhead based on labor hours (and to decentralize production decisions).

Effect of overhead allocation on behavior when inputs are not substitutable.



It is only when the local manager has no flexibility on production inputs that overhead allocations are benign.

For example, we centralize the decision about optimal mix of inputs for production. But now local manager's specific knowledge is not utilized.

Should we centralize the hiring/HR function for the organization's sales team?

Assume each business unit is empowered to hire their own sales team.

However, economies of scale in hiring/HR means that central organization retains HR functions and so incurs all *indirect* support costs related to sales personnel.

- Direct costs are naturally assigned to each business unit.
- How to allocate the indirect/common costs of sales teams?

Crucial; economies of scale in HR (and Finance dept) function. Other tacit knowledge includes; labor regulation, past case expertise, broad experiences, department uptime.

After a certain number (C) of salesperson, per unit cost (marginal cost) may actually increase. Means; there is an optimal number of sales person base on the current phase of the organisation.

- Example; if i am the BU hiring the number C th sales person and if i am charged the MC , i will be unhappy (why am i paying more than other BU).
- this is also inappropriate to measure during BU leader performance appraisal. (even though it is perfectly correct in terms of economics and cost accounting).
- Solutions: consider to outsource some variable HR functions.

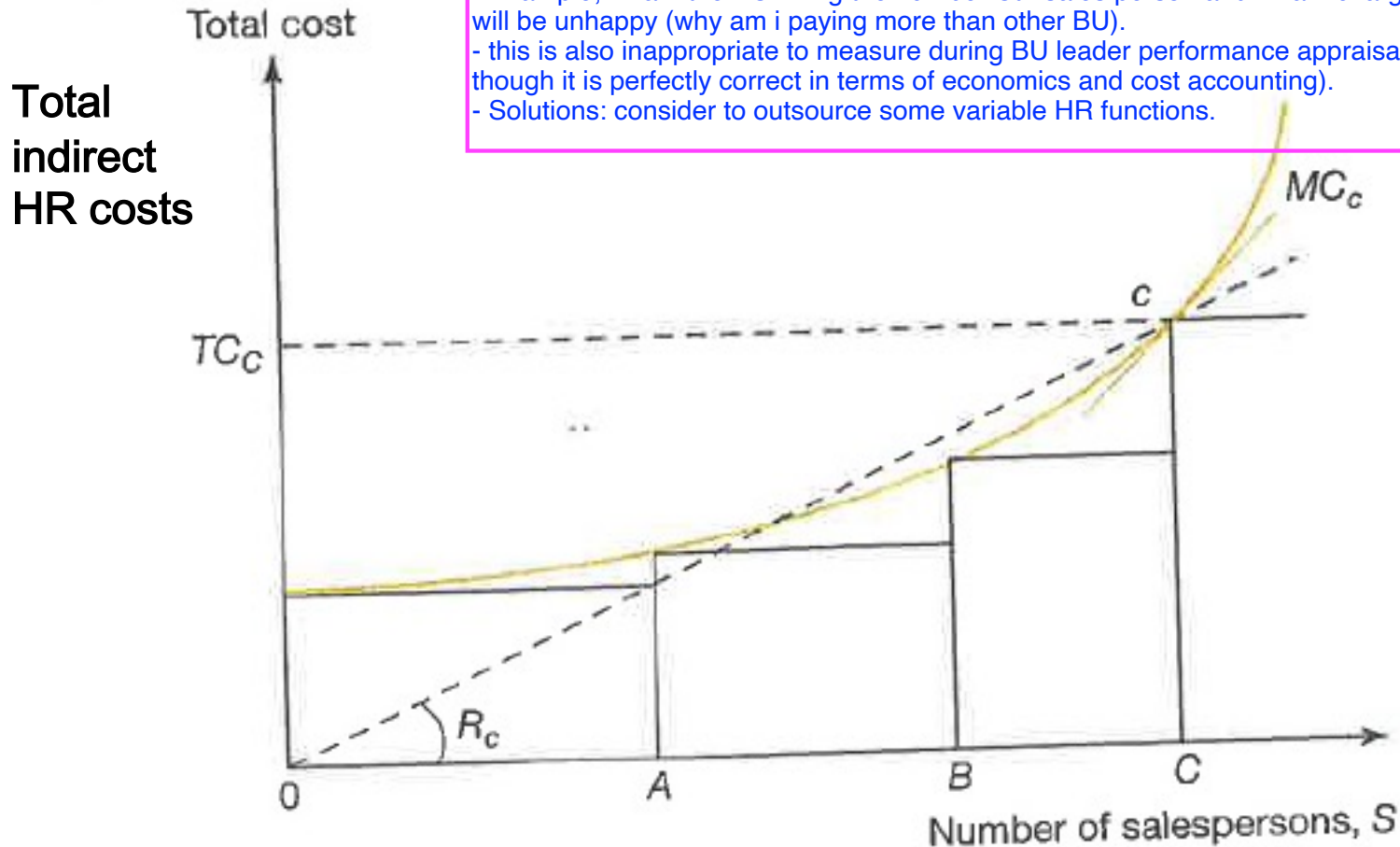


Figure 7.2 from Zimmerman

- TC is the total cost of the centralized HR services necessary to support the sales team – a common (indirect) cost.
- Actual costs incurred (as captured by the accounting system) are a step function because adding HR capacity is “lumpy” (you can’t hire fractions of people).
- If the TC curve is convex, MC increases as we add sales personnel ($MC = \text{slope of the TC curve}$). But it’s impractical to charge (or measure) MC.

- To approximate, we allocate cost based on:
$$\text{Total Cost}/(\# \text{ of sales personnel}) = \text{AC}.$$
- The “right” decision occurs if $\text{AC} = \text{MC}$ so business unit managers equate marginal benefit to MC.
- If $\text{AC} < \text{MC}$, then business unit does not pay full MC but does pay *something* and so reduces consumption relative to what would happen if we did not allocate the joint costs.
- If $\text{AC} > \text{MC}$, price is “too high” and we discourage use of the resource.

So what do you do? Allocate some indirect costs to unit but not “too much” – perhaps allocate those indirect costs we can measure (HR, IT) and understand that this excludes hard to measure costs (more people means longer lines at the café).

Three **necessary elements** for this to work (rules of the game):

1. Set up units as profit centers – managers’ performance evaluation tied to revenues and costs of sales team performance.
2. Assign decision rights over sales hiring to unit managers.
3. Allocate relevant indirect costs (as we have discussed here).

Given complications of allocating indirect overhead/support costs to business units, perhaps **use a *contribution margin* approach** (i.e., revenues – variable direct costs)?

Two main issues:

- How does organization **decide how much to invest in the support function** (i.e., its size)?
- How does organization **decide how to allocate these scarce resources**?

For example, how large a team do we need at Booth to provide administrative support for faculty? **Effectively we “price” these services through the RATS budget**, and so arrive (approximately) at a **market solution**.

Activity-based Costing (ABC).

SBUX Margins/Common Size Income Statements																								
\$M																								

Item 6. Selected Financial Data.

Whole Foods Market, Inc.

Summary Financial Information

(In millions, except per share amounts and operating data)

The following selected financial data are derived from the Company's consolidated financial statements and should be read in conjunction with "Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations" and "Item 8. Financial Statements and Supplementary Data" of this report.

	Sept. 25, 2016	Sept. 27, 2015	Sept. 28, 2014	Sept. 29, 2013	Sept. 30, 2012
Consolidated Statements of Operations Data ⁽¹⁾					
Sales	\$ 15,724	\$ 15,389	\$ 14,194	\$ 12,917	\$ 11,699
Cost of goods sold and occupancy costs	10,313	9,973	9,150	8,288	7,543
Gross profit	5,411	5,416	5,044	4,629	4,156
Selling, general and administrative expenses	4,477	4,472	4,032	3,682	3,355
Pre-opening expenses	64	67	67	52	47
Relocation, store closure and lease termination costs	13	16	11	12	10
Operating income	857	861	934	883	744
Interest expense	(41)	—	—	—	
Investment and other income	11	17	12	11	8
Income before income taxes	827	878	946	894	752
Provision for income taxes	320	342	367	343	286
Net income	\$ 507	\$ 536	\$ 579	\$ 551	\$ 466

Whole Foods Markets, Inc.

\$M

	F2016		F2015		F2014		F2013		F2012	
Sales	15,724		15,389		14,194		12,917		11,699	
COGS	10,313		9,973		9,150		8,288		7,543	
Gross Profit	5,411	34.4%	5,416	35.2%	5,044	35.5%	4,629	35.8%	4,156	35.5%
SG&A Expenses	4,477	28.5%	4,472	29.1%	4,032	28.4%	3,682	28.5%	3,355	28.7%
Other Operating Exper	77		83		78		64		57	
Operating Income	857	5.5%	861	5.6%	934	6.6%	883	6.8%	744	6.4%
Net Income	507	3.2%	536	3.5%	579	4.1%	551	4.3%	466	4.0%

Source: Company F2016 10-K filing.

Operating margin drops from 6% to 5%. Why? has to use ABC to determine why.

If gross margin had been sustained at 35.5% for F2016, would have contributed another \$171M in operating income – critical to understand why gross margin has declined as top-line has grown – higher fixed costs? super-variable costs?

Why ABC?

- Traditional cost systems – which were primarily built to manage manufacturing organizations – do not provide the information necessary for management decisions, including pricing, product mix, logistics, which markets to target, etc.
- Why? They don't capture the big buckets of costs or help us understand what causes those costs to vary.
- This was evident from VMD case (relative importance of overhead costs increased over time; change in nature of activities/costs).

Where are the costs?

Look at Apple, Microsoft, Amazon, Meta – where are the big buckets of costs and **what causes them to vary**, both **across different activities/product lines** and **over time**

	Year Ended December 31,		
	2024	2023	2022
Revenue	\$ 164,501	\$ 134,902	\$ 116,609
Costs and expenses:			
Cost of revenue	30,161	25,959	25,249
Research and development	43,873	38,483	35,338
Marketing and sales	11,347	12,301	15,262
General and administrative	9,740	11,408	11,816
Total costs and expenses	95,121	88,151	87,665
Income from operations	69,380	46,751	28,944
Interest and other income (expense), net	1,283	677	(125)
Income before provision for income taxes	70,663	47,428	28,819
Provision for income taxes	8,303	8,330	5,619
Net income	\$ 62,360	\$ 39,098	\$ 23,200
Cash flows from investing activities			
Purchases of property and equipment	(37,256)	(27,045)	(31,186)
Purchases of marketable securities	(25,542)	(2,982)	(9,626)
Sales and maturities of marketable securities	15,789	6,184	13,158
Acquisitions of businesses and intangible assets	(270)	(629)	(1,312)
Other investing activities	129	(23)	(4)
Net cash used in investing activities	(47,150)	(24,495)	(28,970)

Meta has large buckets of cost that are not directly product-related but critical for its business model (especially R&D and Capex, now including AI spend).

Strategy; depreciate the assets (such as data centers) over a longer time to increase profit in short term. Yet; massive investments are made and how to ensure these companies make proper choices?

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To manage the business, these costs need to be driven to the different processes and activities that consume resources

META PLATFORMS, INC.
CONSOLIDATED BALANCE SHEETS
(In millions, except number of shares and par value)

	December 31,	
	2024	2023
Assets		
Current assets:		
Cash and cash equivalents	\$ 43,889	\$ 41,862
Marketable securities	33,926	23,541
Accounts receivable, net	16,994	16,169
Prepaid expenses and other current assets	5,236	3,793
Total current assets	100,045	85,365
Non-marketable equity securities	6,070	6,141
Property and equipment, net	121,346	96,587
Operating lease right-of-use assets	14,922	13,294
Goodwill	20,654	20,654
Other assets	13,017	7,582
Total assets	\$ 276,054	\$ 229,623

META's biggest (GAAP balance sheet) assets are:

Cash and MS	\$78B
PP&E	\$121B
Goodwill	\$21B

Market cap currently
\$1.96T versus BV of
\$182B (as of 9.18.25).

→ *No brand, no IP, no intellectual capital*

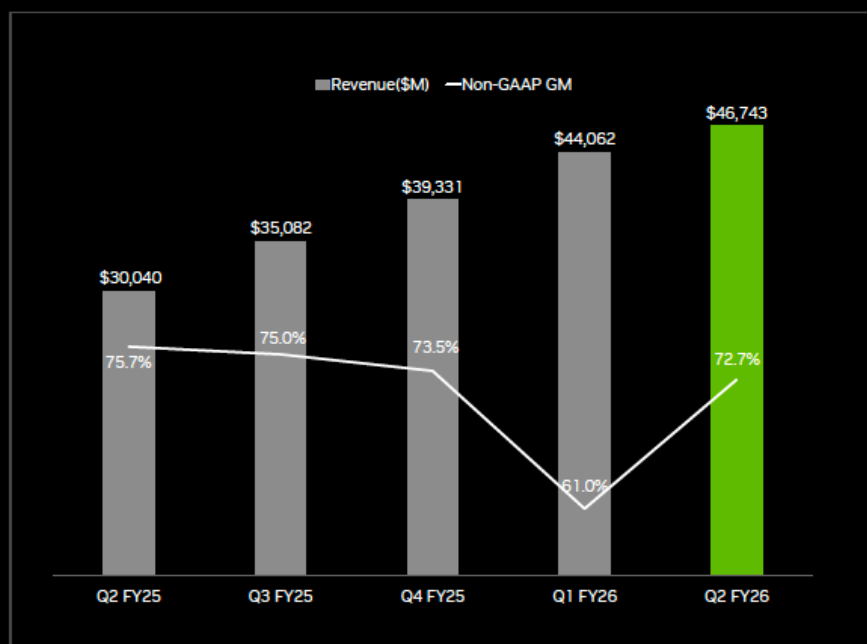
Market cap. vs. book value

Ticker	Market Cap. \$B (8.22.25 close)	Book value \$B (last FQE)
AAPL (Apple)	\$3,380	\$66
NVDA (Nvidia)	\$4,300	\$84
AMZN (Amazon)	\$2,400	\$334
TSLA (Tesla)	\$1,100	\$77
MCD (McDonalds)	\$224	-3
KO (Coca Cola)	\$302	\$29
F (Ford)	\$47	\$45
JPM (JP Morgan Chase)	\$815	\$356
C (Citibank)	\$175	\$213

Q; is Nvidia a crazy bubble?

Main delta between market cap vs book value; the belief that the company will grow its future book value

Q2 FY26 Financial Summary



	GAAP			Non-GAAP		
	Q2 FY26	Y/Y	Q/Q	Q2 FY26	Y/Y	Q/Q
Revenue	\$46,743	+56%	+6%	\$46,743	+56%	+6%
Gross Margin (GM)	72.4%	-2.7 pts	+11.9 pts	72.7%	-3.0 pts	+11.7 pts
GM excluding H2O related charges/ releases, net				72.3%		+1.0 pt
Operating Income	\$28,440	+53%	+31%	\$30,165	+51%	+30%
Net Income	\$26,422	+59%	+41%	\$25,783	+52%	+30%
Diluted EPS	\$1.08	+61%	+42%	\$1.05	+54%	+30%
Cash Flow from Ops	\$15,365	+6%	-44%	\$15,365	+6%	-44%

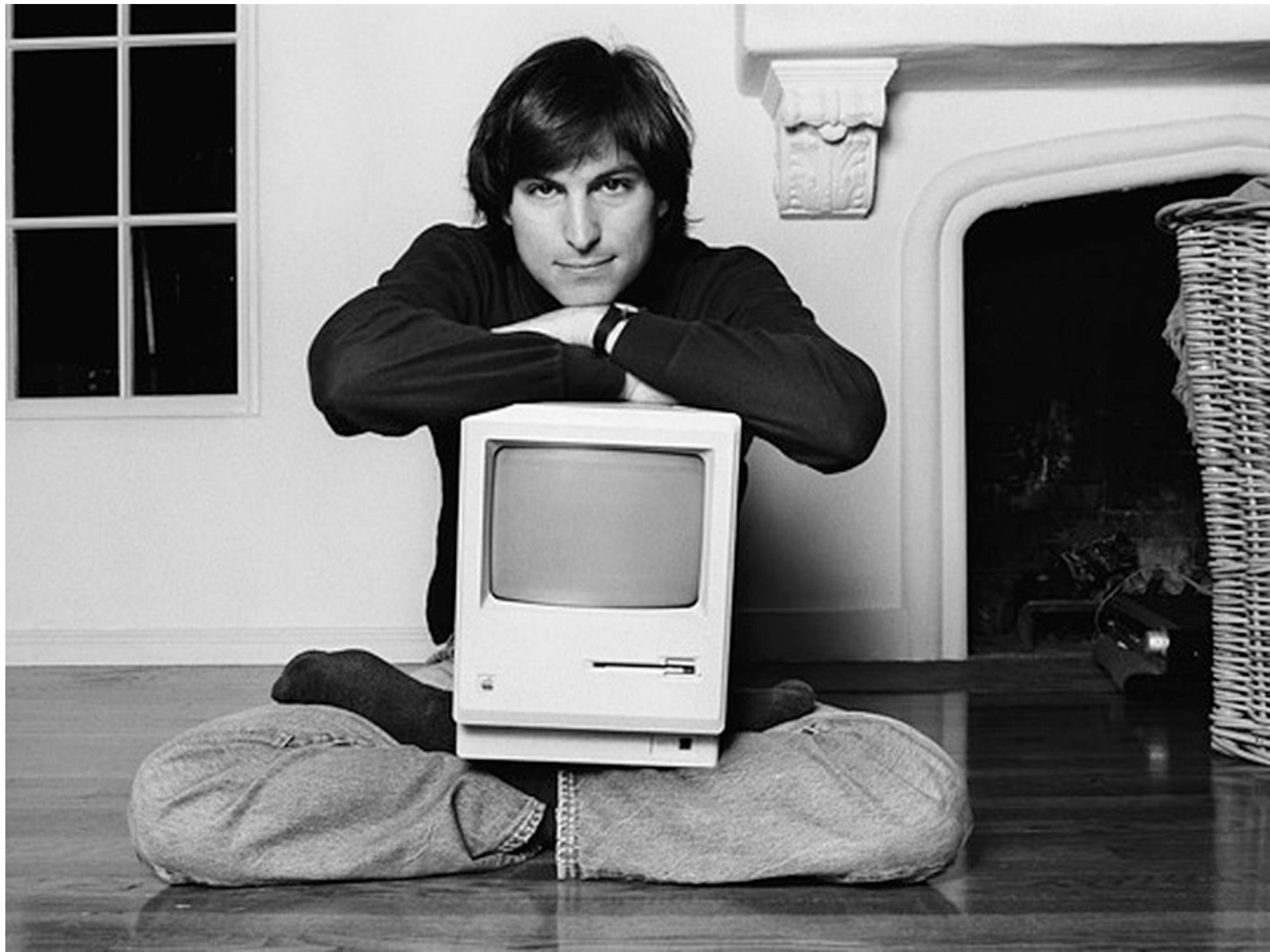
All dollar figures are in millions other than EPS. Refer to Appendix for reconciliation of Non-GAAP measures.

NVIDIA QUARTERLY REVENUE TREND

REVENUE BY MARKETS

(\$ in millions)	Q2 FY26	Q1 FY26	Q4 FY25	Q3 FY25	Q2 FY25	Q1 FY25	Q4 FY24	Q3 FY24
Data Center	\$41,096	\$39,112	\$35,580	\$30,771	\$26,272	\$22,563	\$18,404	\$14,514
Gaming	4,287	3,763	2,544	3,279	2,880	2,647	2,865	2,856
Professional Visualization	601	509	511	486	454	427	463	416
Auto	586	567	570	449	346	329	281	261
OEM & Other	173	111	126	97	88	78	90	73
TOTAL	\$46,743	\$44,062	\$39,331	\$35,082	\$30,040	\$26,044	\$22,103	\$18,120





Plant I produces
only blue pens

Plant II
produces all
of these
colors and
more!



Pen Factory (Cooper and Kaplan):

Factory	Simple (Plant I)	Complex (Plant II)
Annual output:	1,000,000 pens.	1,000,000 pens.
Product types	Blue pen.	Many different types: 100K blue, 60K black, 10K green, + others with volumes as low as 500 (lavender).
Total direct labor hours	Same	Same
Total direct materials	Same	Same
Total machine hours	Same	Same
Fixed costs (property taxes, security costs)	Same	Same
Plant-wide indirect production and distribution costs	Low	High (more indirect production staff, more orders, more purchasing, greater inventory complexity, setup time, quality assurance, etc.)

Pen Factory: Two distortions in Plant II:

1. The reported unit cost of blue pens produced in Plant II is higher than for blue pens produced in Plant I – even though the pens are identical – because indirect overhead in Plant II is substantially higher (and allocated based on # of pens).
2. The cost system reports that all pens produced in Plant II have same unit cost, regardless of volume or the complexity of processes required for the different varieties.
 - This is also a clear distortion—lavender pens produced in low volumes are more costly than blue pens because they consume a lot of overhead resources (e.g., purchases, inventory, set-up time, shipping special orders).

Pen Factory example

Assuming Plant I is efficient, its costs will drive market prices, so margins for blue pens in Plant II will be relatively low (same price but higher unit cost).

Conversely, if the price of lavender pens is higher than for blue pens, reported margins for lavender pens will be higher.

→ *Managers of Plant II will be incentivized to produce an even larger variety of pens at low volume, which increases overhead further, lowering margins and making blue pens seem even less competitive → death spiral.*

Pen Factory example

The problem—which ABC is designed to solve—is that the cost system does not provide the right information. Here, what managers of Plant II really need to know is the cost of the **large diversity of products** they offer.

→ The real driver of costs is variety of products (pen types), which drives (indirect) costs of machine set-up, inventory and storage, ordering, purchasing, distribution, etc.

ABC

Activity-based costing is based on three key ideas:

- The usual allocation bases (DL costs, machine hours) are based on conventional manufacturing models, which are less relevant with automated processes.
- Most organization costs are driven by **activities** unrelated to production/manufacturing – such as **marketing, logistics,** and **R&D/innovation** – and these costs should affect strategic decisions about pricing, product mix, and how the entity allocates its resources.
- The right **cost object** may not be units of output – it may be a batch or a brand or a customer demographic.

ABC systems are designed to answer:

1. What activities and processes consume the organization's resources?
2. What are the costs of these activities and processes?
3. Why is the organization engaging in these activities and processes?
4. How much of each activity or process is required for the organization's products, services, and customers (outputs)?

For Tesla, some key strategic questions are:

- How many different models should we offer?
- Where should we offer those products? Focus on US or go for global presence?
- What types of product support should we offer (charging stations, service facilities, showrooms)?
- Should we focus on automotive business or move into energy storage as well?

ABC provides cost data to help answer these questions.

Step 1. Develop the Activity Dictionary

- Identify activities and processes.
- Described by verbs and objects – *schedule* production, *move* materials, *improve* products, *develop* new products, *respond* to customers.
- Broader than the usual set of cost objects.

similar to our TAA concept. But how micro do we go?

Step 2. Determine cost of activities/processes.

- Obtain cost data from the entity's cost system and drive costs to activities by identifying resource cost drivers.
- For example, costs associated with developing new products are **driven to that activity** (as opposed to being separately recorded in conventional expense categories such as salaries, utilities, depreciation, etc.).
- **Cost drivers** are underlying variables that explain what causes costs to vary (e.g., time or distance may explain variation in the cost of servicing clients); ideally these are causal relationships.

Can think of these as a hierarchy:

- **Unit-level activities:** performed for each unit. Will vary with production or sales volume.
- **Batch-level activities:** performed for each batch or set-up of production; will vary with product variety.
- **Product- or customer-sustaining activities:** Performed for each product and customer.

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In the Pen Factory, the costs of unit-level activities were captured, but those of batch and product-sustaining activities—which drive indirect costs—were not.

ABC systems capture **all** costs – manufacturing, logistics, marketing, R&D – necessary to support these activities.

Step 3. Identify products, services, and customers

- These are the entity's outputs – brands, products, and customers – which become the cost objects.
- Part of the process should include evaluating how much value these generate, so we can make strategic decisions based on the value-added by various products, services, and customers.

Step 4. Identify activity cost drivers

- Activity cost drivers link the costs of the activities identified in Step 2 to the outputs identified in Step 3.
- These may be activity measures like production volume, or time-based measures, or time and intensity-based measures (e.g., time taken x number of high-end personnel required to implement a complex process).
- Measure output of processes/activities as *practical capacity*.

Where to look:

- Look for areas where there are **large chunks of costs** – start with the Income Statement.
- Look for resource consumption that increases with **diversity/complexity** of products, processes, or customers.
- Look for resources for which consumption not well explained by conventional drivers, like units produced.
- Look over periods of years, not months, to identify **longer-run, underlying trends** in cost structure as business evolves.

Should all costs be driven to products, services, customers, etc.?

The answer is (likely) **no** for:

1. **Costs of unused/excess capacity**. As we have already discussed.
2. R&D costs. Costs of product **development** allocated to products or product lines. But **costs of research should be treated as investments** (capitalized as assets for internal management purposes).
 - A similar argument can be made for marketing costs, which drive the value of **brands**.

Transfer Prices (for Birch Paper).

What is a transfer price?

- The way transfer prices are set is part of the organization's "rules of the game." Transfer prices will be most important when profit centers are used – transfer prices determine revenues (COGS) for the selling (buying) unit.
- Transfer prices are for internal transactions and do not affect entity's overall economics.
- Only transactions with entities outside the organization have **direct** economic effects for the entity as a whole.

What is a transfer price?

- Depending on how units are organized (as profit centers, cost centers, etc.), transfer prices can affect unit managers' decisions about transacting with external entities, and therefore have **indirect** economic effects on overall entity.

What is a transfer price?

Price at which goods and services are transferred between business units **within** organizations. Can set at:

- Variable cost (of transferor).
- Full cost (of transferor).
- Full cost + profit margin (of transferor).
- External market price.
- Internal price negotiated by business units.
- Dual price (exit price for transferor is different from entry price for transferee).

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